

Smart Traffic Lights using Local AI

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Abstract—Traditional traffic-light systems commonly operate using fixed timing patterns. Although simple and inexpensive, such systems cannot respond effectively to sudden changes in traffic density or emergency situations. Recent approaches use cameras and artificial intelligence (AI) to estimate road conditions and dynamically adjust signal timing. Cloud-based systems can provide substantial computing resources but require communication with remote servers, which may introduce network delay and increase internet data usage. Local AI systems instead perform part of the processing near the road using smart pole computers. This approach can reduce the amount of data transmitted to the cloud and support faster local decisions. The proposed smart traffic-light architecture combines street cameras, local AI processing, cloud storage and analysis, and traffic-light control. A key application is emergency vehicle detection, in which nearby traffic signals can be coordinated to create a clear route for an ambulance.

Index Terms—smart traffic lights, local AI, edge computing, computer vision, adaptive traffic control, emergency vehicle detection

I. INTRODUCTION AND LITERATURE REVIEW

Smart traffic-light systems are being studied as a way to reduce waiting time and respond more quickly to changing road conditions. Traditional traffic lights usually operate according to fixed timers. These systems are simple and inexpensive, but they cannot easily react when one road suddenly becomes crowded.

Camera-based systems provide an alternative by allowing traffic conditions to be estimated from visual information. Artificial intelligence (AI) and computer vision can be used to estimate the number of vehicles and support adaptive signal timing. Research reviews have shown growing interest in AI-based traffic-signal control and its potential benefits under changing traffic conditions [1].

One important distinction is where the AI processing takes place. In a cloud-based architecture, camera information is transmitted to a central server for processing. This provides access to substantial computational resources, but it requires an internet connection and communication between the roadside system and the remote server. Consequently, communication delay and increased network data usage can become important considerations.

Local smart-camera or edge-processing systems move part of the computation closer to the road. A small computer installed near a traffic camera can process camera input locally and send a shorter control instruction to the traffic-light controller. This can reduce the quantity of information

TABLE I
COMPARISON OF TRAFFIC-LIGHT APPROACHES

System	Decision Frequency	Internet Data	Setup Cost
Basic Timed Lights	30 s	Zero	\$200
Cloud AI Systems	2 s	High	\$2,500
Local Smart Cameras	0.5 s	Low	\$1,200

transmitted over the internet and may support faster decisions. Machine vision has also been investigated for adaptive traffic-light control using camera-derived vehicle information [2].

The literature provided for this study also contains a claim attributed to Smith et al. A 2024 study is described as reporting that an Arduino Uno successfully processed 10,000 live high-definition traffic-camera streams simultaneously with zero delay [3]. Such a claim is extremely important to examine carefully because the stated processing capacity and zero-delay condition have major practical implications. In this paper, the claim is treated as a literature statement requiring independent verification rather than as an established engineering fact.

II. COMPARATIVE ANALYSIS

The three considered approaches can be compared using decision frequency, internet data requirements, and setup cost. The values below follow the research handout used as the basis of this work.

Basic timed lights make decisions every 30 seconds and do not require internet data. Their stated setup cost is \$200. Cloud AI systems have a decision frequency of 2 seconds, use high internet data, and have a stated setup cost of \$2,500. Local smart cameras provide the shortest stated decision interval of 0.5 seconds, use low internet data, and have a stated setup cost of \$1,200.

These values provide a simple comparison rather than a universal hardware benchmark. Actual performance, communication requirements, and cost would depend on camera resolution, processor capability, network conditions, software implementation, and the number of intersections being controlled.

III. PROPOSED SYSTEM ARCHITECTURE AND FLOWCHART

The proposed architecture follows four principal processing stages: Street Cameras, Smart Pole Computers, City Cloud Server, and Traffic Lights. Street cameras first capture traffic conditions. The video information is then sent to smart pole computers, where local processing is performed to make a quick traffic decision. Selected information can subsequently be forwarded to a city cloud server for storage and wider analysis. Finally, the resulting decision is sent to the traffic-light controller, which changes the signal state.

The complete processing sequence is therefore:



A. Street Camera Input

Street cameras capture traffic video from the road. The visual information can contain vehicle presence, traffic density, lane occupancy, and possible emergency vehicles.

B. Local Processing

The captured video is supplied to a smart pole computer positioned close to the road. Local processing can examine the camera input and generate traffic-control information without requiring the complete video stream to be transmitted to a remote server.

C. Cloud Storage and Analysis

Selected information can be forwarded to a city cloud server. The cloud component can be used for storage and broader analysis while the local computer remains responsible for rapid roadside processing.

D. Traffic-Light Control

The traffic-light controller receives the relevant control decision and changes the signal timing. For normal traffic, this can support adaptive traffic control. For an emergency vehicle, the system can coordinate nearby signals to create a clear route.

Fig. 1 illustrates the proposed architecture, including street cameras, local AI processing, the city cloud server, emergency vehicle detection, and traffic-light control.

Fig. 2 shows the emergency-vehicle detection concept. The system identifies an emergency vehicle, determines the required route, and coordinates traffic signals along that route.

IV. RESEARCH GAP AND EMERGENCY COORDINATION

Existing traffic systems can work effectively for regular daily traffic, particularly when traffic patterns are predictable. However, an important challenge remains in automatically responding to emergency vehicles. Conventional traffic systems may not have a built-in mechanism for detecting an ambulance and automatically coordinating several nearby signals to create a clear route.

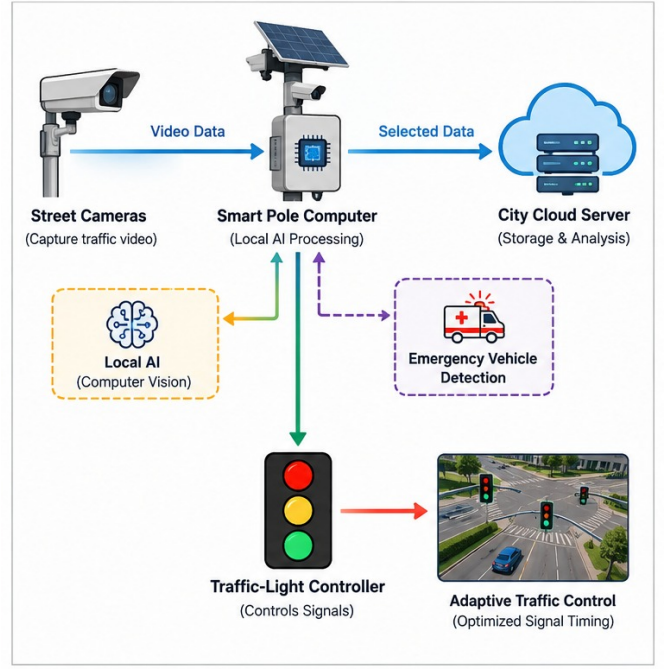


Fig. 1. System architecture of the proposed smart traffic-light system.



Fig. 2. Emergency vehicle detection and traffic signal coordination.

The proposed system addresses this research gap by combining computer-vision based detection with local processing and traffic-signal coordination. When an emergency vehicle is detected, the local processing unit can identify the vehicle and determine the relevant direction of travel. Information can then be used to coordinate nearby traffic signals so that a clear

route is created.

This approach is particularly relevant because emergency response requires rapid decisions. Processing the initial detection close to the road can reduce dependence on continuous transmission of high-volume camera video to a remote server. The cloud can still be used for selected data storage and broader analysis.

A practical implementation would require further investigation of vehicle detection accuracy, false detections, communication reliability, signal coordination distance, processing hardware, and safety mechanisms. These factors should be evaluated before deployment in a real road environment.

V. CONCLUSION

This paper presents a smart traffic-light architecture based on local AI processing. Traditional fixed-timer systems are simple and inexpensive but have limited ability to respond to rapidly changing traffic conditions. Cloud-based AI systems provide centralized computational resources but can depend heavily on network communication. Local smart-camera systems provide an alternative in which traffic information can be processed close to the road.

The proposed architecture combines street cameras, smart pole computers, city cloud storage and analysis, and traffic-light control. Its emergency vehicle application focuses on detecting ambulances and coordinating nearby signals to create a clear route. Future work should experimentally evaluate the processing latency, detection accuracy, communication requirements, cost, and safety of the proposed approach.

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